

3(a)(i)	t_3 and t_7
(a)(ii)	t_4 and t_8
(b)(i)	Frequency of oscillation $f = \frac{4200}{60} = 70 \text{ Hz}$
(b)(ii)	Maximum acceleration $a = \omega^2 x_0$ $= (2\pi f)^2 x_0$ $= (2\pi \times 70)^2 (2.5 \times 10^{-2})^2$ $= 4.8 \times 10^3 \text{ m s}^{-2}$

